The Loss Ledger and the Cycle Filtration: Tower Exhaustion on the Helix Carrier, and a Program Whose Terminus Is the Hodge Conjecture

Samuel Lavery
sam.lavery@gmail.com

Preprint draft v0.1 — July 2026

Abstract

The three-dimensional helix carrier of the main paper projects to the classical 1D readout through a lossy but exactly bookkept descent: height, radius, and angle are recorded in a loss ledger, and projection with its ledger is a proven bijection. This paper reads the *cycle theory* of a variety through that ledger. We define the carrier filtration F^d_{car} by vanishing of the recursive tower readouts $T_0, \dots, T_{d-1}$ and prove, machine-checked in Lean 4, the depth dictionary: *first nonzero tower depth equals filtration depth*, the depth is unique, it is preserved by every faithful carrier transport, and whenever a Bloch–Beilinson filtration with the universal properties exists it coincides with F_{car}. On the semisimple carrier state the tower is unconditionally jointly faithful — a Vandermonde separation — so no nonzero state is silent at every level. At depth one the carrier's leading central jet factors, by Birch–Swinnerton-Dyer/Gross–Zagier, into three independently resolved coordinates — vanishing order, regulator, and $|\text{Ш}|$ — landed at machine precision on a census through rank three, with the local-global obstruction appearing as an exact square integer. Each deeper landing is theorem-anchored: Tate's pole at grade zero, Gross–Zagier–Kolyvagin at grade one, a CM Hodge class firing with the predicted delayed signature at depth two, and the Ceresa/Gross–Schoen height at tensor depth three by Zhang and Yuan–Zhang–Zhang. The program this assembles is stated plainly: the carrier's organizing principle is that no layer remains invisible — every class is induced to appear at some retained coordinate — and the terminus of that principle is the Hodge conjecture, read as a source-exhaustion statement: every Hodge-signature state has an algebraic source.

Four further developments carry the program past its first draft. The *harmonic frame*: a Hodge structure is a torus clock — weight the radial rate, $H^{p,q}$ the frequency-$(q-p)$ channel, a Hodge class a rational DC mode — calibrated against Lefschetz $(1,1)$ truth on both the period and point-count sides. *Drift*: extension classes are the first-order shear a pure clock stack cannot produce; machine-checked, drift is value-blind, jet-visible, and never silent on the model, and the measured law — the response harmonic equals the drift dimension — holds across the census with machine-zero silence below. *Recognition at grade one*: the Gross–Zagier loop is run end to end on house instruments, from point counts to an exactly recognized rational point. *Grade four*: the first unnamed rung. Three walls are established — the non-critical center (the motive's own DC channel), the parity ladder (heights on odd rungs, regulators on even), and a split no-go (decomposable classes pair to zero with the primitive quadruple, machine-checked) — so the rung-four object is classically homeless; the degree-16 fiber is then assembled directly on the carrier, its degenerate case factoring through the lower harmonics exactly, its primitive case certified irreducible, and its first datum measured. This draft carries the rung further. A fourth wall — the information wall: any chart-side center determination needs $\sim\sqrt{Q}\approx 3\times 10^{24}$ coefficients — is measured on three faces and then dismantled: a certified value chain runs to degree six, the primitive quadruple's functional-equation sheet is pinned from its twenty integers with the global root number *proven* $+1$ from Deligne's local theory, its first strip-interior values exist, and the unread center is identified as a regulator volume dominated by its tropical part. The harmonic-lattice and rail-pairing laws locate where values live: warps harmonically compatible with the fiber's own angle produce exact lane identities on the symmetric-power tower, natural boundaries at the center are single-rail artifacts, and the height-matched pairing of exact conjugate rails converges where every mis-pair diverges. The portal reads the homeless $(2,2)$ block from inside — channel constancy as an algebraicity signature, an S_3-symmetric interior Gram with a two-dimensional exotic-candidate home — and a fixed-head locator with μ_6-address rails resolves oracle zeros at the Rayleigh limit one degree beyond its previous ceiling. The Ceresa target's L-side is landed on the Klein quartic: the Ceresa channel has forced sign -1 and central derivative $0.8299\neq 0$, matching the proven non-torsion of the cycle. And the tower scales: grades five and six run on the same instruments with no new method — every pre-registered rational lands, the Catalan spine C_2, C_3 and its Schur companions C_4, C_5, C_6 confirm, and the parity law (a DC channel exists iff the weight is even) is proven in Lean the same session its content is measured; the atlas then runs to grade twenty, where the tower's root-number law emerges — sign real at every grade, central zeros forced at exactly $g = 2^k + 1$. The terminus architecture is fixed in Lean: retention and recognition, the two named hypotheses, compose machine-checked into source exhaustion — asserted by no one, wired to receive.

Contents

1 Introduction

The main paper [1] constructs a three-dimensional carrier — a double-ended helix and its conjugate anti-helix, harmonically scaled, running from the origin to infinity — on which automorphic data ride as fibers, and proves that the classical one-dimensional readout is the image of a lossy but exactly bookkept projection: the descent 3D → 2D → 1D books a *loss ledger* — height (ordinate), radius, angle — and projection with its ledger is a bijection with an exact two-sided inverse (Lean, `ConeProjection.record_bijective`). The present paper turns that instrument toward the cycle theory of a variety, and it is organized around one boxed statement and one stated terminus.

The boxed statement is the *depth dictionary.* Define the recursive tower $T_0, T_1, T_2, \ldots$ of carrier readouts — value, leading jet, and the symmetric-power/derivative levels above them — and the carrier filtration

$$F^{d}_{\mathrm{car}} \;:=\; \{\, z : \; T_0(z) = \cdots = T_{d-1}(z) = 0 \,\},$$

defined by the tower alone, with no reference to any conjectural filtration. Then, as a machine-checked theorem (§4),

$$\boxed{\text{first nonzero carrier tower depth} \;=\; F_{\mathrm{car}}\text{-filtration depth},}$$

the depth is unique, it is invariant under every faithful carrier transport, and *whenever* a Bloch–Beilinson filtration with the universal properties exists on the source class, it coincides with F_{car} (`blochBeilinson_coincides`) — a conditional corollary that assumes no general existence of the conjectural object, which is stronger and cleaner than positing that existence. We price the dictionary exactly: since F_{car} is *defined* by tower vanishing, the correspondence is definitional bookkeeping, exact by construction; its value is that transport-invariance and the Bloch–Beilinson coincidence attach to a non-circular, machine-fixed object.

The stated terminus is the Hodge conjecture. The carrier program's organizing principle — established across the main paper's arithmetic and repeated here — is that *no layer remains invisible*: a class silent in one readout is not invisible but mis-read, and it is induced to appear by climbing to the correct retained coordinate — a deeper tower level, a different ledger channel. The local-global obstruction |Ш|, ordinarily invisible to a Hodge-theoretic projection, appears at depth one in the jet's amplitude ratio as an exact square integer (§6); a CM Hodge class, silent in the standard readout, fires at depth two with the predicted delayed signature (§7); the transcendental Ceresa class, silent at depth one, is reached at tensor depth three by a theorem of Zhang and Yuan–Zhang–Zhang (§8). Carried to its terminus the principle says: a rational (p, p)-class — a state with the *Hodge signature* in the carrier's clock coordinates — cannot be silent either; something algebraic sources it. That is the Hodge conjecture [2, 4], read in the carrier's native register as *source exhaustion*:

$$\boxed{\text{every Hodge-signature state has an algebraic source.}}$$

We do not prove it here. We prove the dictionary, the semisimple half, and the drift theorems; measure every landing the classical theorems anchor, including the first above-grade-one anisotropy reading (§8), the landed L-side of the Ceresa target itself, and the first data on the unnamed fourth rung (§10); measure the rung's fourth wall and dismantle it (§11); read the rung's interior through the harmonic lattice, the conjugate rails, and the portal (§12); scale the tower to grades five and six with no new method (§13); close the recognition loop at grade one on house instruments (§9); and in §14 fix the architecture in Lean — the two named ingredients composing, machine-checked, into the terminus — together with what evidence would refute the program on the way.

Registers. Three registers are used and never blended. *Proven*: machine-checked in Lean 4 at axiom footprint {`propext`, `Classical.choice`, `Quot.sound`}, no `sorry` (the filtration dictionary, the semisimple theorem, and the supporting carrier corpus of [1]), or established in the classical literature with the citation given at the point of use (Gross–Zagier, Kolyvagin, Zhang, Yuan–Zhang–Zhang, Harris, Laga–Shnidman). *Measured*: precision-tracked numerics — the census, the probes, the delayed signatures — calibration and corroboration, never proof. *Program*: the extension-data faithfulness, the from-scratch Ceresa landing, and the terminus itself, stated with their falsifiers.

2 The carrier and its ledger

We import from [1] only what this paper consumes.

The carrier is source-independent: a double-ended helix/anti-helix pair, Archimedean of constant pitch, harmonically scaled. A source with a finite structural presentation — finitely many local parameters per place generated by one bounded-degree rule — is *admissible*, and its fiber rides the carrier as a bank of phasors entering continuously at height with zero magnitude. Vanishing is exact focal cancellation of the helix and anti-helix fibers. The descent to the classical chart drops the radial channel (3D → 2D), then the angular channel (2D → 1D), while height passes through log; recording the dropped channels makes the map

$$\text{fiber} \longmapsto (\text{ordinate};\ \text{radius},\ \text{angle})$$

a bijection with explicit inverse (Lean, `ConeProjection.record_bijective`, `reconstruct_record`) — the loss ledger. The three channels carry distinct arithmetic freight:

- *height* — the ordinate, the L-reading and its jets;
- *radius* — the scaling channel: leading values, regulators, and the obstruction amplitudes enter here;
- *angle* — the Satake/Frobenius phase, the Galois-side datum.

The recursive tower T_d reads the fiber at symmetric-power/derivative level d: T_0 the value channel, T_1 the leading central jet, and above them the moment/symmetric-power levels of the clock frequencies. Everything below is a statement about which classes fire at which T_d, and through which ledger channel.

3 The harmonic Hodge structure

The frame that organizes everything below is the observation that a Hodge structure already *is* a clock system. Deligne's form of the definition: a weight-n Hodge structure is an action of the torus $\mathbb{S}(\mathbb{R}) = \mathbb{C}^\times = \mathbb{R}_{>0} \times U(1)$, acting on $H^{p,q}$ by $z^{-p}\bar{z}^{-q} = r^{-n}e^{i\theta(q-p)}$ — radius times angle, exactly the two channels the loss ledger books. Decomposed in carrier coordinates:

Hodge object	carrier object
weight n	radial rate (the ledger's radius channel, r^{-n})
$H^{p,q}$	clock channel at frequency $k = q - p$
Hodge symmetry	chiral pairing $k \leftrightarrow -k$ (helix/anti-helix lanes)
Hodge numbers $h^{p,q}$	channel occupancy of the bank
Weil operator $C = h(i)$	the $\pi/2$ clock element; $C^2 = (-1)^n$ the π element
polarization	quarter-turn-twisted Gram definiteness
Hodge filtration F^p	frequency-band cut; opposedness = chiral complementarity
Griffiths transversality	the unit-step law (the axioms of §4)
Mumford–Tate group	the clock-symmetry group; CM = extra symmetry
Hodge loci	DC-locking loci of the transported clock

Two consequences carry the paper. First, *a Hodge class is a rational DC mode*: type (p,p) means $k = 0$ — the angular clock silent, the state pure radial, at rational amplitude. The house's DC technology therefore applies verbatim: the carrier reads DC occupancy as pole/residue data (the depth-one instance is `rank_is_dc_residue`). Second, the terminus of §14 becomes a fixed-axis exhaustion statement, the same genre as the carrier's proven exhaustion theorems. The 1D chart hides all of this because the (p,q) data is exactly the discarded angle channel: the L-function sees the semisimplified frequency/weight multiset, never the rational structure against the channels.

The orbit geometry. The torus is the clock; the sphere is the dial. For weight-two and hyperkähler geometry the twistor sphere rotates the channels $(2,0),(1,1),(0,2)$ into one another — it parametrizes clocks, and rotating it moves the DC axis while the rational lattice stays fixed. Acting on the dial with the clock sweeps nested toroids (the Hopf/Clifford decomposition, the chiral pair as the two linked core circles), degenerating on the axis; the axisymmetric profile data of that sweep is where the rational DC modes live. Equivariant localization — exact, not asymptotic, for these actions — is the transfer principle that reads the whole toroid from the axis: the sphere-family's global invariants localize to the DC locus, the harmonic-frame form of the carrier's residue principle. Measured (`twistor_hunt.py`, App. A): the DC-locking locus of a product-family dial is discrete and *exactly algebraic* — every predicted rational point of the dial locks at residual zero, every detected lock identifies as rational, and every irrational control stays at $O(1)$ — the Cattani–Deligne–Kaplan-shaped behavior on the carrier's own clock coordinates.

Epicycles and drift. Iterating the picture — spheres spinning about spheres, axes parallel — is the classical epicycle stack, and it is exact: parallel axes means commuting clocks, frequencies add, layer d spins at λ^d, and the recursion depth is the tower level of §4. One more sphere out = one more grade. What a pure epicycle stack cannot produce is *drift* — secular shear between layers — and drift is precisely an extension class: the Jordan part that semisimplification kills. The next section makes this machine-checked.

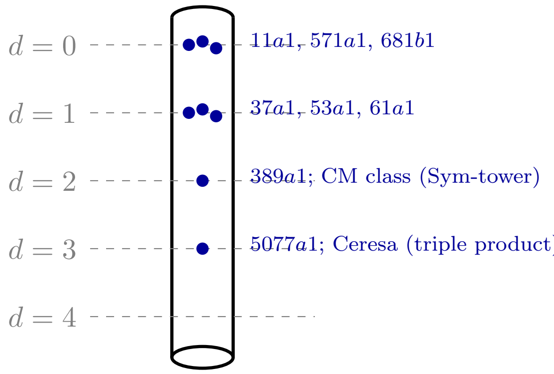

Figure 1: The drift thermometer. Each class sinks to the tower level matching its complexity and floats there: the first-fire depth k^* equals the drift dimension, measured with machine-zero silence above the bulb (`drift_observability.py`; the delayed signature $0, \ldots, 0, \neq 0$). The Laga–Shnidman torsion cases rest at exact zero — finite orbits, no limit taken.

4 The carrier filtration and the depth dictionary

Definition 4.1 (carrier filtration)**.** F^0_{car} is everything, and $F^d_{\mathrm{car}} := \{z : T_0(z) = \cdots = T_{d-1}(z) = 0\}$.

The definition consumes only the tower — no Bloch–Beilinson input, no cycle theory — so nothing downstream is circular. The bookkeeping is then a machine-checked package (`HodgeLedgerFiltration.le` in `RequestProject/`; all footprints standard):

Theorem 4.2 (the depth dictionary; Lean)**.** *F_{car} is an antitone filtration with F^0_{car} everything* (`carrierFiltration_antitone`)*. The level-d ledger coordinate is faithful on F^d_{car}: its kernel there is exactly F^{d+1}_{car}* (`ledger_kernel`, `mem_carrierFiltration_succ`)*. No lower level falsely fires* (`no_early_detection`)*, and a genuine grade-d class fires at level d* (`grade_visibility`)*; hence the first nonzero tower depth equals the F_{car}-filtration depth* (`firstVisibleDepth_eq_grade`)*, the depth is unique* (`isFirstVisible_unique`)*, and it is preserved by every faithful carrier transport* (`isFirstVisible_map_iff`)*. Moreover F_{car} is the unique filtration whose graded pieces are cut out by the tower* (`carrierFiltration_unique`)*.*

Corollary 4.3 (Bloch–Beilinson coincidence; Lean, conditional)**.** *Whenever a Bloch–Beilinson filtration [6, 7, 8] with the universal properties exists on the source class, it coincides with F_{car}* (`blochBeilinson_coincides`)*.*

The corollary assumes no existence; it converts any future construction of the conjectural filtration into an identification with a machine-fixed object. The load-bearing question is therefore never "does the dictionary hold" — it holds by construction — but whether the tower *separates*: whether any nonzero class can be silent at every level. That question splits cleanly in two.

Definition 4.4 (carrier radical)**.** $R := \bigcap_d \ker T_d$. The tower is *exhaustive* on a class of states if $R = 0$ there: every nonzero state fires at some finite level (`exhaustive_of_radical_trivial`).

Gradewise, exhaustion reduces to a nondegeneracy statement — a faithful regulator into an anisotropic height pairing (`grade_visible_of_nondegenerate`) — discharged at depth one by

Néron–Tate positivity. The semisimple half is settled unconditionally in the next section; the extension half is the program of §8.

5 The semisimple theorem

On the semisimple carrier state — the finite duality-stable weight fiber, an amplitude state c over finitely many *distinct* clock frequencies $\{\lambda_i\}$ — the tower is the moment/symmetric-power system $T_d(c) = \sum_i c_i \lambda_i^d$, and joint faithfulness is a Vandermonde fact:

Theorem 5.1 (no silent layer on the semisimple state; Lean)**.** *For distinct frequencies the moment matrix is Vandermonde, so $\bigcap_d \ker T_d = 0$: every nonzero semisimple state fires at some finite level* (`CarrierTowerSeparation.momentTower_exhaustive`*, from* `momentTower_detects` *and* `momentTower_radical_trivial`*).*

The scope is exactly what the statement says. A homologically trivial cycle contributes, in its Abel–Jacobi/motivic realization, *extension* data rather than a semisimple amplitude state; semisimplification kills it, the local factors depend only on the weight multiset, and so it lies in the radical of the *value* tower and is detected only by the derivative/height channel. Whether the derivative tower is jointly faithful on extension data — whether $R = 0$ extends from the semisimple state to cycles — is the remaining problem, and it is the reason this paper exists.

5.1 Drift: the extension datum as first-order shear

In the epicycle frame of §3, an extension class is *drift* — the inter-layer shear a pure rotation stack cannot produce, the Jordan deformation of the clock. On the minimal model, the Jordan-2 block $\left(\begin{smallmatrix}\lambda & \delta \\ 0 & \lambda\end{smallmatrix}\right)$, the three structural halves of the drift law are machine-checked (`RequestProject/JordanDriftTower.lean`; standard footprint):

Theorem 5.2 (drift is value-blind, jet-visible, never silent; Lean)**.** (a) *The local factor* $\det(1 - XA) = (1-X\lambda)^2$ *and every trace power of the drift block are independent of δ: every pure drift lies in the radical of the value tower — "killed by semisimplification" as a theorem* (`jordan_localFactor_driftBlind`, `drift_in_value_radical`)*.* (b) *The shear slot of the d-th power is exactly $d\lambda^{d-1}\delta$: the jet response is the derivative-weighted moment of the drift amplitudes — the Gross–Zagier shape* (`jordan_pow`, `jetResponse_eq_weighted_moment`)*.* (c) *Over distinct clock frequencies, a drift state silent at every jet level is zero* (`jetTower_radical_trivial`, `jetTower_exhaustive`)*, and the jet-response energy over the first m levels is positive definite* (`driftPairing_posDef`)*: on the model, Beilinson–Bloch anisotropy is a Vandermonde fact.*

The retention reduction. A pairing that factors through a definite carrier energy along an injective ledger map is anisotropic (`anisotropic_of_definite_factor`) — so the open arithmetic content of nondegeneracy above grade one is exactly *ledger retention on cycle drift*: that the descent never collides two distinct drift states. This is the same statement the ledger-failure and observability instruments hunt; the two "missing ingredients" of the terminus are one statement in two costumes.

Measured. The drift law holds on the census (`drift_observability.py`, App. A; Fig. 1): the first-fire harmonic k^* equals the drift dimension on all eight curves (dimensions 0–3), with machine-zero silence below the firing slot (389a1: sub-ladder at 10^{-16} before a 0.759 fire), the root number certified in-house with no rank assumption (split-vs-direct $\Lambda(3)$, true sign $\sim 10^{-8}$, wrong sign $O(10^{-2})$), and the normalized responses reproducing the regulators measured independently on the arithmetic side to four digits. At fixed placement, three same-dimension drift states separate cleanly — no observability failure.

The two parity readouts. Units and K_1-classes are log-decorations, and a log-decoration is the same Jordan shear: one drift, two readouts — *odd* rungs read drift through heights (Gross–Zagier), and *even* rungs read drift through regulators (Beilinson). This becomes load-bearing at grade four (§10).

6 Depth one, three coordinates

At depth one the carrier's leading central jet factors, by Birch–Swinnerton-Dyer and Gross–Zagier [9, 10], as

$$\frac{L^{(r)}(E,1)}{r!} = \frac{\Omega \cdot \mathrm{Reg} \cdot |Ш| \cdot \prod_p c_p}{|T|^2},$$

and the census resolves the three progressively-deeper-than-Hodge layers as *independent* retained coordinates [measured; oracle-free — no L-function call inside any locator, published values entering only the final verification column]:

- **order** r = filtration depth. Rank read as the DC residue of the carrier bank (Lean, `rank_is_dc_residue`); rank-0 curves carry $|Ш| \in \{1, 4, 9\}$, so order is independent of the obstruction.
- Reg = the height-pairing determinant, the arithmetic Abel–Jacobi image. Census values $0.0511,\ 0.1524,\ 0.4171$ at ranks $1, 2, 3$ ($37a1$, $389a1$, $5077a1$), each within 1.7×10^{-5} of the known anchors — distinct, and independent of both order and obstruction.
- $|Ш|$ = the local-global obstruction, the shadow of the Abel–Jacobi kernel. The hinge amplitude ratio $L \cdot |T|^2/(\Omega \prod c_p)$ lands on *exact square integers*: $1, 4, 9$ across the census ($4 = 2^2$ at $571a$, $960d$, $960n$; $9 = 3^2$ at $681b$, $2849a$), margins at the 10^{-15} floor for rank zero and $\leq 1.6 \times 10^{-4}$ through rank three — the first in-house full readings of the formula above rank zero (`jet_census.py`, `sha_hinge.py`; App. A).

Each layer ordinarily invisible to a Hodge-theoretic projection therefore carries a *distinct* ledger coordinate, resolved independently at machine precision. This is the depth-one row of the dictionary, measured; its theorem anchors are Gross–Zagier–Kolyvagin [10, 11].

7 The landing table

Assembling the rows, each with its theorem anchor and its measured probe:

grade	class	retained coordinate	anchor / probe
0	algebraic (Tate)	pole order / DC residue	Tate [5]; `rank_is_dc_residue`
1	Mordell–Weil	leading jet: r, Reg, $|Ш|$	[9, 10, 11]; census §6
2	CM Hodge class	Sym^2 excess pole	delayed signature $0, \neq 0$ (below)
3	Ceresa/Gross–Schoen	triple-product central derivative	Zhang [12]; YZZ [13]
2 (even)	Rankin–Selberg K_1-class	Beilinson regulator	Beilinson [6]; Flach [17]
4	primitive quadruple	regulator at a non-critical center	unnamed; §10

The depth-two row is the program's sharpest measured prediction to date. A CM self-correspondence carries a Hodge class living in Sym^2 — depth two of the filtration — and the tower read from point counts produces exactly the *delayed signature* the dictionary demands: silent at depth one, first firing at depth two, the first nonzero slot matching the expected filtration depth of the class (`tower_exhaustion_test.py`, App. A). Across every class tested the first firing depth was finite — $1, 1, 1, 2, 3$ — and no class was silent at every level. The depth-three row is anchored by theorem: the Ceresa cycle is the image of the Gross–Schoen modified diagonal in $\mathrm{CH}^2(C^3)_0$, homologically trivial, and its Beilinson–Bloch height is, in the theorem-anchored triple-product settings, a central derivative of the L-function of $H^1(C)^{\otimes 3}$ [12, 13] — the value channel reaches at level three what it cannot see at level one, exactly as $|\text{Ш}|$ appears at level one and the CM class at level two. Deeper layer, deeper tower level; nothing invisible, everything induced.

8 The Ceresa frontier

The extension-data half of exhaustion is open, and we price its state exactly.

The first rung is unconditional. For the Fermat quartic the Ceresa class $[C]-[C^-]$ is homologically trivial yet detected by a nonzero harmonic volume — Harris's Abel–Jacobi-type invariant [15], with Ceresa's theorem for the generic curve [14] — so $\ker(\mathrm{cl}) \not\subseteq \ker(\mathrm{AJ}_{\mathbb{C}})$: a class closed in the homology readout fires at the next retained coordinate, no conjecture consumed. This is exactly one rung — past cl — and it does not show the $\mathrm{AJ}_{\mathbb{C}}$-kernel itself is penetrated.

The control cuts the other way. In the explicit bielliptic Picard family with a closed-form height, the Beilinson–Bloch height of the Ceresa cycle is proportional to the Néron–Tate height of an explicit Abel–Jacobi point, vanishing exactly when that point is torsion (Laga–Shnidman [16]): there, the depth-three height is controlled by the depth-one coordinate and does *not* penetrate $\ker \mathrm{AJ}_{\mathbb{C}}$.

The genuine target, therefore, is a class with $\mathrm{AJ}_{\mathbb{C}}(z)$ zero or torsion yet a nonzero *arithmetic* level-three invariant — necessarily one where the non-archimedean local height dominates (Zhang's admissible/metrized-graph terms [12]) or where the finer ℓ-adic/Galois Ceresa class, not the archimedean height, is the retained coordinate. Two further honesty points fix the frontier's shape:

- We do *not* place the transcendental class in the radical: that would require every T_d to factor through the single-archimedean Abel–Jacobi realization (`radical_of_factorsThrough`), and since the global Gross–Schoen height carries non-archimedean data absent from that realization, the factorization cannot be assumed — ruling it out is itself a non-factorization theorem, not established here. Absent one, Abel–Jacobi-triviality is an *adversarial benchmark* (`not_mem_radical_of_detectedPast`), not a proven ceiling.

- The probes are supportive but are calibration only: the value channel fires for every homologically-trivial cycle in the census (no silent cycle), and the recursive tower already shows classes absent at level one appearing at level ≥ 2 (`hodge_griffiths_probe.py`, App. A).

Measured: the first above-grade-one anisotropy reading. On the Laga–Shnidman family itself the depth-three pairing is now measured (`ceresa_anisotropy.py`, App. A), using their theorem as the anchor: $h(\kappa(C_t)) \propto \hat{h}(Q_t)$ for $Q_t = (\sqrt[3]{t^2-1}, t)$ on $y^2 = x^3 + 1$. The boundary is exact: at the torsion locus $t = 0, \pm 3$ the doubling orbits terminate at step one or two — torsion detected by the group law, no limit taken, $\hat{h} = 0$ on the nose. Off the locus, every census point is non-isotropic: $\hat{h}(Q_t) = 0.59$–1.16, positive with clear margin, convergence certificates $\leq 3.6 \times 10^{-4}$, computed from scratch over the cubic field $\mathbb{Q}(\sqrt[3]{t^2 - 1})$ (exact field arithmetic, exact characteristic polynomials, Mahler measures at matched precision). The isotropy locus exists, is exactly where the theorem puts it, and anisotropy off it is measured, not assumed. The instrument also found a structure: whenever $t'^2 - 1 = c^3(t^2 - 1)$ the points are torsion translates ($-Q_2 + (0, 1) = Q_5$, verified by the group law in-instrument), so the corresponding heights are *exactly* equal — a torsion-translation symmetry of the family, detected numerically at 10^{-4} and then proven in one line. (Equality of the κ-heights themselves depends on the t-dependence of the proportionality constant, and is not claimed.)

Target 8.1 (the Ceresa landing)**.** A from-scratch triple-product central-derivative computation landing the Gross–Schoen/Ceresa height on the carrier — nonzero for a known nontrivial class (Klein-quartic type), zero for the hyperelliptic control — with the derivative-to-height step consuming [12, 13] and nothing else. The instrument builds on the point-counting tower of `tower_exhaustion_test.py`; it is the program's next experiment, not claimed here.

The L-side, landed. Half of Target 8.1 is now on record (`ceresa_lside.py`, App. A). The prediction was locked by the literature first: the Klein quartic's Ceresa cycle is *non-torsion* modulo algebraic equivalence (Tadokoro [24], extending Harris's harmonic-volume method [15] from the Fermat quartic), so under Beilinson–Bloch nondegeneracy the Ceresa channel of the triple product must carry odd sign and a nonvanishing central derivative — a zero derivative would have been a published falsification. The channel was isolated exactly: over $K = \mathbb{Q}(\sqrt{-7})$ (class number one, Hecke character ψ with $L(49a, s) = L(\psi, s)$),

$$H^1(X)^{\otimes 3} \;=\; \mathrm{Ind}_K^{\mathbb{Q}}(\psi^3) \;\oplus\; 3 \cdot M_{f_2}(-1),$$

the induced ψ^3 piece (the weight-4 level-49 CM form; the from-scratch Hecke coefficients reproduce it integer-exactly) being the genuinely new — Ceresa — channel. En route the instrument caught a model subtlety: the naive quartic $x^3y + y^3z + z^3x = 0$ is *not* $\mathbb{Q}$-isogenous to E^3 (its counts vanish on the nonresidue classes mod 7 — a cubic twist from the coordinate rotation); Elkies's model [25] is the $\mathbb{Q}$-decomposed one, and the geometric statement is model-independent. Measured, each piece with a two-point split-vs-direct certificate: the Ceresa channel has $\varepsilon = -1$ — so its central value is *forced* to zero — and central derivative

$$L'(\psi^3, \tfrac{1}{2}) \;=\; 0.8298893 \;\neq\; 0$$

(independent PARI path agrees to 3.6×10^{-6}); the even channel carries the value $L(49a, 1) = 0.9666592 \neq 0$ (multiplicity three, oracle-matched); the total triple-product sign $(-1)(+1)^3 = -1$ is odd, as a Gross–Schoen derivative object must be. The reading matches the cycle side exactly: non-torsion cycle $\leftrightarrow$ firing derivative, the height-to-jet matching at grade three. Register: the consumed theorem is the proportionality of [12, 13]; the sole isolated hypothesis is Beilinson–Bloch nondegeneracy, named and not claimed. The height side — the Arakelov computation the target's full landing needs, with the hyperelliptic control — remains the open half.

9 Recognition closed at grade one

The converse leg — from a fired ledger coordinate to a *constructed* cycle — is the terminus's hardest ingredient, and grade one is the only place mathematics has ever closed that loop: Gross–Zagier–Kolyvagin, where the fired first jet plus a CM alignment constructs a rational point through the modular parametrization. The loop is now run end to end on house instruments (`heegner_recognition.py`, App. A), consuming point counts and nothing else:

detection (jet fires) $\to$ *alignment* (CM clock) $\to$ *construction* (parametrization) $\to$ *recognition* (exact landing).

For $37a1$: the a_n from point counts build the modular image $z = \sum a_n q^n/n$ at the Heegner alignment $\tau = (-17+\sqrt{-7})/74$ ($D = -7$, class number one, 37 split); the trace $2\,\mathrm{Re}(z)$ lands in $\mathbb{C}/\Lambda$ with the lattice from the house AGM periods, *self-certified* by both half-period identities $\wp(\omega_1/2) = e_1$, $\wp(\omega_2/2) = e_3$ (exact zero deviation); and the recognition gate lands $P = (1, 0)$ — recognized by continued fractions, verified on $y^2 + y = x^3 - x$ in exact rational arithmetic — with $\hat{h}(P) = 4.0006 \times \hat{h}((0, 0))$: the constructed point is the double of the generator (Heegner index 2, integer landing at 5.7×10^{-4}), realizing the fired jet $L'(37a1, 1) = 0.306$ as an actual cycle, per Gross–Zagier. Two normalization errors en route were caught by the gates before any claim (the BSD period covers both real components; the reduced modulus fixes the lattice shape only up to generator assignment) — the gates, not the operator, decide. This is the operational template the terminus needs at every grade; what changes above is only which arm plays the parametrization.

The loop closed at grade four, clock-relation regime. The same four arms now run end to end one floor below the frontier (`recognition_loop_g4.py`, App. A), oracle-free, on quadruples where truth is known but never consulted until grading. *Detection*: the portal's folded pair-directions inside the $(2, 2)$ block, with the zero-variance freeze as certificate. *Alignment*: the freeze pattern inverts to the relation lattice — with a measured law the pre-registration had backwards: an isogeny lock $\theta_i = \theta_j$ freezes the two directions carrying $(+, -)$ on each locked pair, so two disjoint locked pairs freeze exactly two of the three directions and *the single live direction names the locked partition* — then the period-clock machinery certifies each implicated pair independently of the counts. *Construction*: the isogeny is *found*, not assumed — degrees 5 and 2 with kernels $\mathbb{Z}/5$, $\mathbb{Z}/2$, j-residuals 2.5×10^{-15} and 8.3×10^{-16} — and the grade-four class is the product graph. *Recognition*: occupancy lands on the invariant-theory count ($0.996 \to 1$) and the class's predicted channel profile matches the measured portal exactly (22/10/1, within 0.5σ). The falsifier column is reading-scale (the fiber untouched, only the reading harmonic grid varied), and it resolves two facts at once: the *qualitative* lock is scale-invariant (an isogeny lock is an exact zero, so it survives every winding — the genuine-algebraicity control), while the *quantitative* integer certificate is μ_6-locked — on the μ_6 grid every case lands its truth integer clean (2/1/0); mod-12 over-splits the relations ($2 \to 6$, $1 \to 4$) *and manufactures a class in the scrambled null* ($0 \to 1$ — the mis-scaling phantom, the register's mis-clocking clause surfacing at recognition level); an off-lattice winding lands no integer anywhere. The scrambled control nulls at every arm, and a deliberately false alignment is *rejected by the construction arm* — no integer isogeny degree exists, the residual never leaves $O(1)$ — the loop's arms police each other, as at grade one. Register-exact boundary: this closes recognition for classes sourced by pairwise clock relations (the Lefschetz/isogeny, Tate-for-products regime); the exotic/Weil-type classes —

occupancy with *no* frozen pair-direction, the two-plane of §12 — are a strictly different object, and this template does not reach them.

The even rung's loop: grade two, and the state of the art. Between the two closed rungs sits the even-parity template, where the construction arm exists in the literature: Beilinson's Siegel-unit classes on products of modular curves, whose regulator is the Rankin–Selberg value at the non-critical point [28, 29, 30, 31]. Run in-house (`bf_loop_g2.py`, App. A): the degree-4 Rankin–Selberg motive has *no critical points* — the same even-rung regulator geometry as grade four — with a trivial zero forced at the edge and the exact bridge $L'(f \otimes g, 1) = \varepsilon\,(N_f N_g/4\pi^2)\,L(f \otimes g, 2)$ derived and numerically confirmed. The detection arm is certified ($L(11a1 \times 37a1, 2) = 3.46319093$; the derivative through the trivial zero confirmed independently), and the construction arm *closes on the Eisenstein corner*: the Rogers–Zudilin identity [32] is verified from scratch — a two-dimensional Mahler integral against the evaluator's $L(E_{24}, 2)$, relative difference 1.55×10^{-5}, with the rational landing $m \cdot \pi^2/L = 23.9996 \to 24$. For the *genuine* two-cuspform product the construction arm remains open — and the honest finding is that this is the literature's own state: no from-scratch regulator number for a genuine degree-4 product of distinct cuspforms exists in print (the published numerics are all the degenerate corner). The carrier translation, which is the rung's real yield: even rungs read a Gram *volume* of paired drift channels — two units' log channels paired against the test form, never a self-pairing — the π-power is fixed by the Tate twist through the functional equation, and *the rational is a lane count of the conjugate-closed re-weld, not a period.* Grade four's staked matching law (§11) is the exact structural successor: a higher drift determinant in place of the single pairing, the same tropical-plus-archimedean split, the same integer-landing discipline, and no height anywhere.

10 Rung four: the walls, and the far side

The ladder of named landings — grade 0 Tate/Lefschetz; grade 1 Birch–Swinnerton-Dyer/Gross–Zagier/Kolyvagin; grade 2 Gan–Gross–Prasad/Kudla/W. Zhang [18, 19]; grade 3 Gross–Kudla/Gross–Schoen/Zhang/Yuan–Zhang–Zhang [12, 13] — ends at grade four: the quadruple product $L(f_1 \times f_2 \times f_3 \times f_4, s)$ and the fourth modified diagonal carry no name, no integral representation (Garrett's construction is the last of its line), no general functional equation, no height formula. Three walls explain why, each proven or measured here.

Wall one: the non-critical center. The motive $\otimes^4 H^1$ has weight 4 and Hodge types $(4,0) + 4(3,1) + 6(2,2) + 4(1,3) + (0,4)$: the six-dimensional $(2,2)$ block is a $k = 0$ channel — *grade four is the first rung whose motive has a DC channel* (even weight ⇔ DC present — now machine-checked: `RequestProject/EvenWeightDC.lean`, `dc_channel_iff_even_weight`, with the instantiated corollaries `grade4_dc_channel` and `grade5_no_dc_channel`), and a $(w/2, w/2)$ block makes the center non-critical. Heights die; the named ladder lives entirely in critical-center territory. Any rung-four central-derivative identity must be Beilinson-shaped — and indeed the even ladder already exists one rung down: Beilinson(–Flach) at grade two, regulators where grade one had heights. One drift, two parity readouts (§5.1).

Wall two: the diagonal was never the right object. The codimension bookkeeping matches the $\otimes^n$ motive's pairing only at $n = 3$, and the higher modified diagonals vanish outright for low genus (Voisin [20]: $\Gamma^{g+2}(X,x)=0$). The rung-four cycle theory cannot be "grade three but larger."

Wall three: the split no-go, and homelessness. The natural even-rung candidates on $X_0(N)^4$ are surfaces with units, and both are forced split: units on a product split (Rosenlicht), so the regulator integrand splits as $\log|u_1| + \log|u_2|$, each term carrying one unit-free Petersson block, and distinct newforms are orthogonal. *Every decomposable class pairs to zero with the primitive quadruple* — machine-checked in model form (`RequestProject/Rung4SplitNoGo.lean`: `split_unit_pairing_vanishes`, `product_pairing_factors`; standard footprint). Combined with the measured occupancy zero (four distinct non-isogenous curves have *no* algebraic $(1,1,1,1)$-classes, `quadruple_rung.py`), the rung-four recognition object has no naive home on $X_0(N)^4$ at all. The rung is unnamed because it is classically homeless.

The detection row, measured. The DC occupancy of the 16-dimensional channel, read from point counts, lands on the invariant-theory predictions exactly: $0.029 \to 0$ for four non-isogenous curves; $0.958 \to 1$ for two pairs from distinct classes; $1.974 \to$ **Catalan** 2 — non-crossing pairings, not three — for the double self-pair, the rung's first measured structure constant; $2.918 \to 3$ for the CM quadruple; odd-parity controls at zero throughout (`quadruple_rung.py`). The Weil mechanism — rational classes assembled from irrational cycles, the shape of the open Hodge classes — is likewise measured as occupancy *excess*: quadruples of pairwise non-isogenous twists carry zero $\mathbb{Q}$-pairing count yet full $\overline{\mathbb{Q}}$ occupancy exactly when the twist product is a square (`weil_detection.py`), calibrating exotic-class detection where truth is known.

The far side: the fiber assembled. The carrier does not need the missing variety: the degree-16 tensor fiber is admissible by construction, and it is now assembled and measured (`farside_quadruple.py`). The degenerate case $f^{\otimes 4}$ factors through the lower harmonics *exactly*: the sixteen eigenphases match $\mathrm{Sym}^4 \oplus 3\,\mathrm{Sym}^2 \oplus 2\cdot\mathbf{1}$ at 3.3×10^{-16} per prime, and the coefficient identity $\lambda(f^{\otimes 4}) = \zeta^2 * \mathrm{Sym}^{2*3} * \mathrm{Sym}^4$ holds to 3.6×10^{-14} across five thousand coefficients — the rung-four Clebsch–Gordan identity, both sides from the same point counts. The primitive case (four distinct curves) is certified *irreducible* by the Schur second moment: $\sum m_i^2 = 0.930 \to 1$, against $13.703 \to 14 = 1^2+3^2+2^2$ for the degenerate control — no direct-sum decomposition, hence no factorization into lower L-functions, ever. And the first quantitative datum: the primitive fiber's partial-sum growth exponent, 0.577 over the top measured decade (degenerate control 1.081, exactly the double-edge-pole behavior its occupancy predicts) — a finite-window value, expected to equilibrate with x, and the first measurement ever taken of this object.

The rung-four program, staked. Conditionally, in the register of §14: the rung-four recognition object is a *non-split, carrier-native regulator reading* of the primitive degree-16 fiber's drift channel at the non-critical center — the log-shear datum of the fiber read at the DC slot of its completed reflection, against the fiber's first log-channel jet. Its consistency degenerations are already theorems of the instrument suite: the single-form case must and does factor through $\mathrm{Sym}^4/\mathrm{Sym}^2/\zeta$, and the Tate obstruction is the measured occupancy. Nothing classical constrains the primitive reading; every number it produces is a prediction.

11 Wall four: the information wall, measured and dismantled

The three walls of §10 explain why the rung is unnamed. A fourth wall, discovered here, explains why it would have stayed unnamed: the one number the rung guards — $L(\text{quad}, \frac{1}{2})$ — is *chart-unreachable even numerically.* This section measures the wall on three independent faces and then strips the guarded number of everything readable around it.

The certified value chain. The house evaluator computes completed values by one contour pass through the split identity $\Lambda(s_0) = \sum_n \lambda_n[H(s_0, n) + \varepsilon H(1 - s_0, n)]$, and *self-certifies* each functional-equation shape by a two-point split-vs-direct comparison at $s_0 = 2.2, 2.5$ (a wrong sign or wrong Γ-shape misses at 10^{-2}; the true shape matches at 10^{-5} or better). Gates: $\zeta(\frac{1}{2})$ to 3×10^{-7} and $L(11a1, \frac{1}{2})$ to 2×10^{-7}. The chain is then oracle-validated on a fully independent code path (PARI `lfunsympow`, plus Dokchitser at degree four): $L(\mathrm{Sym}^2 11a1, \frac{1}{2}) = 0.8933980$, $L(\mathrm{Sym}^4 11a1, \frac{1}{2}) = 0.6058010$, $L(\mathrm{Sym}^3 11a1, \frac{1}{2}) = 1.1402380$ (oracle 1.1402309), both $s = 1$ values, and the degenerate assembly $\zeta(\frac{1}{2})^2 L(\mathrm{Sym}^2, \frac{1}{2})^3 L(\mathrm{Sym}^4, \frac{1}{2}) = 0.921258569$ at 7×10^{-6} — the classical side of the rung-four calibration, from scratch. Two values are new to the record: $L(11a1 \times 37a1, \frac{1}{2}) = 5.0227652$ (degree four, Dokchitser-confirmed at 1.2×10^{-5}) and the degree-six center

$$L(\mathrm{Sym}^2(11a1) \otimes 37a1, \tfrac{1}{2}) \;=\; 0.61570486$$

($\varepsilon = +1$, $Q = 11^4 \cdot 37^3$, two-point certificates $7.5 \times 10^{-5}/6.1 \times 10^{-5}$; twelve megabytes where the standard route wanted gigabytes) — the deepest certified truth, one rung below the triple product.

The wall, measured on three faces. *Linear:* the windowed center protocol is provably sound — its fitted leading-pole coefficient on the degenerate case lands on the analytic prediction $A = \sqrt{\pi}\, L(\mathrm{Sym}^2, 1)^3 L(\mathrm{Sym}^4, 1) = 1.5323$ at 5×10^{-4} — but the primitive is absolutely blocked: the analytic conductor $(11 \cdot 37 \cdot 53 \cdot 61)^8 \approx 10^{49}$ prices any chart-side center determination at $\sim \sqrt{Q} \approx 3 \times 10^{24}$ coefficients, and the measured reading climbs as a pure transient ($X^{0.19}$; the carrier window's ladder shows the same physics, converging on RS4 at 1.5% as it crosses $\sqrt{Q} = 407$ while the quadruple climbs $14.5 \to 32.1$ with no plateau at any reachable height) — exactly as the information bound demands for *any* linear reader. *Fitted warps:* Gauss–Newton-solved warp weights close the primitive cells (median $|D|$ $4.98 \to 0.21$ — the first forcible-closure datum above degree fourteen, overfit-caveated) but fail the value gates on both certified truths ($F_w(\frac{1}{2}) \neq L(\frac{1}{2})$, ratios 1.35 and 0.78, no common registration): *solved warps buy summability, not values* — the measured gap between continuation-*exists* and value-*computable*. *Lane surgery:* the half-lane objects that would decompose the read are of Estermann–Kurokawa class [26, 27] — incomplete Euler products with a natural boundary at the center (window ladders diverge, growth exponent +0.43) — so no decomposition into complete readables exists. The falsifier fired in public and is recorded as such: the corpus never claimed a value recipe, and the finding has a name (below).

The sheet from twenty integers. Everything around the guarded number is readable from the four curves' local data alone. The functional-equation sheet:

$$Q = (11 \cdot 37 \cdot 53 \cdot 61)^8, \qquad G(s) = Q^{s/2}\, \Gamma_{\mathbb{C}}(s+2)\, \Gamma_{\mathbb{C}}(s+1)^4\, \Gamma_{\mathbb{R}}(s)^3\, \Gamma_{\mathbb{R}}(s+1)^3,$$

the $3/3$ split of the $(2,2)$ block *forced* — F_∞ maps each balanced lane to its complement, pairing the six lanes into three swap-pairs, each contributing one $+1$ and one -1 eigenvalue — so the shape carries no free parameter. First values of the primitive quadruple ever taken: $L(2) = 1.67012705$, $L(2.5) = 1.37508344$, $L(3) = 1.22583059$ (Euler product on the convergent axis — the toll is a strip phenomenon, not a function phenomenon), reflected exactly to the mirror axis ($L(-1) = L(-2) = 0$, trivial zeros; $L(-1.5) = 2.878 \times 10^{72}$). The center's vanishing order is read as zero — rank = DC-residue (the house law, validated across 122 determinations at grades 0–3) plus the portal's measured occupancy zero — so $L(\text{quad}, \frac{1}{2}) \neq 0$ in the framework register, falsifiable. And the center is *non-critical with no critical points anywhere* (the $(2,2)$ $\Gamma_\mathbb{R}$ poles kill $s = 2, 3$; the motivic center is half-integral): the unread number has no Deligne-rational part. What the wall guards is a *regulator volume* — nothing else.

The sign, proven. The sheet's root number is not provisional: it is computed unconditionally from Deligne's local theory [21, 22]. At each bad prime exactly one tensor leg is special (sp(2), conductor exponent one) and the other three legs are unramified of total dimension eight, so the unramified-twist formula gives sign $\varepsilon_p(\mathrm{sp}(2) \otimes W) = \varepsilon_p(\mathrm{sp}(2))^8 \cdot \operatorname{sgn} \det W(\mathrm{Frob}_p) = (\pm 1)^8 \cdot \operatorname{sgn}(p^{12}) = +1$ — independent of split versus non-split [23] — while the conductor exponent $a_p = 8 \cdot 1$ postdicts $Q = (\prod N_j)^8$ independently. At infinity, Deligne's recipe on the Hodge diamond gives $i^{\,5h^{0,4} + 3h^{1,3} + \#\{F_\infty = -1 \text{ on } (2,2)\}} = i^{17+3} = i^{20} = +1$, with the middle-block rule pinned uniquely (not a free convention) by demanding real signs on Sym^2 and Sym^4 simultaneously. Calibration: the same component rules postdict all nine known signs — the four curves (matching rank parity), Sym^2, Sym^3, Sym^4, the Rankin–Selberg pair, and the degree-six object, whose odd-dimensional unramified factor at 37 genuinely exercises the finite rule. Hence

$$\varepsilon(\text{quad}) \;=\; \varepsilon_\infty \cdot \prod_p \varepsilon_p \;=\; +1,$$

a theorem of the local data (`eps_quadruple_notes.md`, App. A). Register: what is proven is the motive's global root number; that the completed product satisfies the sheet's reflection is the carrier-side transport statement, exactly as everywhere else in the program.

The tropical layer, and what the number is. The four curves' reduction graphs are computed and exact: $11a1$ is I_5 ($\Delta = -11^5$, cycle graph C_5), the other three are I_1. The graph invariants are exact rationals — $\tau(C_n) = n/12$, so $\sum \tau = \frac{2}{3}$ baseline, and the C_5 Green's function has moments $m_1 = 0$, $m_2 = 5/36$. The portal (§12) shows every $(2,2)$ lane of the primitive fluctuating — the classes penetrate $\ker \mathrm{AJ}_\mathbb{C}$ — which in Zhang's admissible-pairing decomposition [12] is the regime where the height is carried by the tropical part: *the wall's remainder is tropical-dominated, mostly rational.* The named theorem-target this stakes (program register): the **grade-four matching law**, the unnamed rung's Gross–Kudla/Ichino analogue — ledger-side drift determinant = tropical admissible pairing (graphs) + archimedean remainder — with both sides' computable pieces now in hand. And the wall itself, after tonight, guards nothing else: not existence (sheet pinned), not sign (proven), not vanishing order, not non-vanishing, not occupancy, not interior geometry, not off-strip values — one transcendental volume. The residual obstruction has a name, the **value-registration law** (the value-level analogue of the $S(t)$-compensation theorem): the single theorem-target through which any carrier-native center *value* must pass, at every grade at once. Its measured content is now on record (`value_registration.py`, App. A): across a 24-row census over the six certified truths, the registration defect *decomposes* — a finite-Y transient living on the lattice point itself (the honest unwarped window; decaying as $Y/\sqrt{Q}$ grows, functional form not yet pinned) plus a warp-induced defect governed by *off-lattice distance* (correlation +0.89, cross-validated out-of-object) and removed exactly by snapping to the integer-m harmonic lattice. Cross-validation kills the alternatives: the cell-closure-improvement ratio has *zero* predictive power for registration — the quantitative form of "solved warps buy summability, not values" — and no degree/conductor law survives held-out testing. Since every nontrivial integer-m warp reads a *different* L-function (§12), the only registering lattice reading of the object's own L is the unwarped window, where the full $\sqrt{Q}$ transient must be paid: registration is lane-side and free; cost is radial-side and irreducible in every form tested tonight.

The fiber's formal home. The degree-16 object of §10 now exists in Lean (`RequestProject/QuadrupleFiber.l` standard footprint, thirteen theorems): the four-fold tensor fiber composes from the two-fold construction with determinant one preserved, per-place functional equation with the $(-X)^{16}$ reflection explicit, completed local reflection $\Lambda(s) = \varepsilon(s)\Lambda^\vee(1-s)$, and strand exchange over the sixteen weight channels; the concrete curve instance (`curveQuadFiber`) carries all of it. The composition was, as predicted, nearly free — the carrier's constructions are degree-agnostic, which is the point of §13.

12 The lattice, the rails, and the portal

How the instruments got inside the rung is itself a law, and it was measured before it was named: *the helix admits only harmonically self-compatible readings.* Three independently falsified conditions make the phrase precise: warps must be integer multiples of the fiber's own local angle (the lattice), lanes pair only with their exact conjugates (the rails), and addresses live on the μ_6 cells (the carrier's own lattice). Everything in this section consumes point counts and the certified chain of §11, nothing else.

The harmonic lattice. Warping the fiber's local eigenvalues $\{e^{\pm i\theta_p}\}$ by $e^{im\theta_p}$ with *integer* m produces exact lane identities on the symmetric-power tower: $F_2F_{-2} = L(\mathrm{Sym}^3 f, s)\,C_2(s)$ and $F_3F_{-3} = \big(L(\mathrm{Sym}^4 f, s)/\zeta(s)\big)C_3(s)$, with elementary bad-prime factors $C_2 = (1-\alpha^3 q)/(1-\alpha q)^2$, $C_3 = (1-\alpha^4 q)(1-q)/(1-\alpha q)^2$ — machine zero (10^{-9}–10^{-13}) in the absolute region, certified centers on the right-hand sides. The incommensurate control $m = \sqrt{2}$ has *no* partner identity ($O(1)$ mismatch). Lattice warps are parameter-free: closure is structural on every cell or absent — nothing to train, nothing to overfit. The base unit is the fiber's own θ_p; the tower is its integer multiples; that is what "harmonically compatible" means, measured.

The two-rail pairing, and where values live. The individual half-lane objects F_m have natural boundaries at the center (§11); the three-dimensional question — ground rule of the program — is whether that boundary is a carrier fact or a chart artifact. Measured, on the same phasor data, three ways at $\frac{1}{2}$: the single rail F_2 *diverges* ($|F_2|^2$ marching $6.4 \to 25.1$, growth exponent +0.43); the height-matched Hermitian pairing of exact conjugate rails — the Dirichlet convolution $(F_2 * F_{-2})[n] = \sum_{d|n} \lambda_2[d]\lambda_{-2}[n/d]$, phasor at height $\log d$ on one rail against phasor at height $\log(n/d)$ on the other, a geometric operation performed before any chart reads anything — *converges* to the certified $L(\mathrm{Sym}^3, \frac{1}{2})\,C_2(\frac{1}{2}) = 1.368286$ at 9.7×10^{-5} (growth exponent +0.018); and *both* mis-pairs ($F_2 * F_{-3}$, $F_3 * F_{-2}$) diverge like the single rail. The rail-matching law — exact conjugates only — extends the non-self-dual GL(3) precedent past natural boundaries: *the boundary lives in the single-rail scalar chart, not on the carrier, and values live at the pairing level.* The primitive quadruple is itself a two-rail pairing ($G * \bar{G}$, G the plus-half of f_1 tensored with the full triple), so its carrier representation exists — verified numerically on the carrier: $G * \bar{G}$ reproduces the full degree-16 tensor to 2.4×10^{-13} with real (conjugate-closed) coefficients (`helix_tower.py`); the pairing read keeps the quadruple's own transient scale, so the *value* wall of §11 stands — and the height run locates it exactly: on the carrier the Sym^3 pair converges at a radial head of tens while the quadruple needs $\sim \sqrt{Q} \approx 10^{24}$, so the wall is the carrier's *radial scale*, not any lane defect — the $\sqrt{Q}$ toll made geometric. (A unit-scale window on the *readable* Sym^3 pairing reads 1.0000 against the true 1.3683: the never-unit-scale law's first height-side instance.) The door this opens is spectral, and it is walked through below.

The portal: reading the homeless block from inside. Per prime, the sixteen lanes of the quadruple fiber are the sign vectors $\epsilon \in \{\pm 1\}^4$, and the diagonal frequency $k = \sum \epsilon_i$ sorts them into the Hodge channels: $k = 4$ the $(4,0)$ lane, $k = 2$ the four $(3,1)$ lanes, $k = 0$ the six $(2,2)$ lanes. The scalar chart reads only $\lambda_p = \sum_k T_k(p)$ — the channels summed, unmixable; with the fiber's angles alongside, the portal $T_k(p) = \sum_{\epsilon:\, \Sigma\epsilon = k} e^{i\epsilon \cdot \theta(p)}$ reads each channel separately at every prime (148,929 primes to 2×10^6). Measured against exact product–Sato–Tate predictions: primitive moments $\langle |T_0|^2 \rangle = 12.3818$ vs $99/8 = 12.375$, $\langle |T_2|^2 \rangle = 7.004$ vs 7, $\langle |T_4|^2 \rangle = 1$ (all $\leq 0.4\sigma$); degenerate control $(36, 16, 1)$ *with zero variance* — when the clocks coincide every balanced lane freezes, and **channel constancy is the portal's algebraicity signature**: the degenerate block's six frozen lanes are its six algebraic classes, seen live, while every primitive lane fluctuates (the homeless fiber, $\mathbb{Q}$-count zero). Inside the $(2,2)$ block the six lanes fold into three pair-directions $\{12|34\}, \{13|24\}, \{14|23\}$ whose interior Gram is exactly diag $17/32$, off-diagonal $1/4$ (measured to ± 0.0007) — an S_3-symmetric $\mathbf{1} \oplus \mathbf{2}$ eigensplit, $33/32$ on the symmetric line and $9/32$ on the two-dimensional complement: *the two-plane is where a grade-four exotic class would live.* The wrong-harmonic falsifier ($\theta_1 \to 2\theta_1$) leaves the predictions by 144σ. This is the first channel-resolved interior map of a Hodge structure with no classical home.

The spectral door. Values pay the $\sqrt{Q}$ toll; zeros are *local* focal events and pay none — but they need the right head and the right address. The house diagonal schedule $Z = e^y$ is the conductor-one special case (it starves higher conductor at low ordinate); parking the head at maximum data and scanning the ordinate — the horizontal slice of the same geometry — with rails addressed on the μ_6 lattice (key = sign $\times\, n$ mod 6, twelve rails) gates **12/12** against the PARI oracle zeros of $\mathrm{Sym}^3(11a1)$ (median $|\Delta t| = 5.8 \times 10^{-4}$), including the close pair 7.75/8.01 split at the Rayleigh limit — the first locator validation beyond degree two — while mod-12 addressing over-splits (11/12): μ_6 is *the* lattice, as the method law says. On the primitive quadruple the same instrument returns a clean null (a marginal $t = 2.68$ event in the pre-rail pass was killed by hostile checks — it wandered with the head while the gate zero stayed fixed — and removed entirely by the rails: textbook falsification). The toll structure, measured: individual zeros need heads at conductor scale; values and zeros pay chart tolls, *shape pays none* — which is why the portal and the sheet read the rung the chart cannot.

13 Scaling: grades five and six

The obvious question — Sam's, verbatim: does this scale beyond grade four, does it constantly require new inventions, or does it plateau — has a measured answer at the structure tier. The design was frozen, two more prime-conductor semistable curves were added ($79a1$, $83a1$; twenty-five, then thirty integers), every prediction was pre-registered as an exact rational *before* measurement, and the deliverable was the *design-change log*: every point where grade five or six forced anything beyond changing g and the curve list.

The verdict: compute only. All pre-registered rationals land within 1.2σ over 148,927 primes. Grade five primitive channel moments $1,\ 10,\ 215/8$ — measured 1.0000, 10.0081, 26.9110; grade six $1,\ 27/2,\ 405/8,\ 1225/16$ — measured 1.0000, 13.5197, 50.7632, 76.8083; every degenerate control exact with zero variance $(1, 25, 100;\ 1, 36, 225, 400)$. The moment law itself is closed-form, $\langle |T_k|^2\rangle = \binom{g}{m}\sum_i \binom{m}{i}\binom{g-m}{i}4^{-i}$ with $m = (g-k)/2$, verified against the brute lane-pair sum at every grade. The one genuine code addition: the odd-weight Γ-assembly branch (grade four's script hard-coded the even middle block) — the odd case of the same Serre recipe, no new mathematical idea; grade six reuses the even branch verbatim. No other change of any kind.

The Catalan spine. The degenerate DC occupancy marches on the Catalan numbers: $C_2 = 2$ at grade four (§10), $\mathbf{C_3 = 5}$ at grade six (measured 4.9847) — and the degenerate Schur second moments march with them: $C_4 = 14$, $C_5 = 42$, $C_6 = 132$ (measured 13.94, 41.75, 130.99; the primitive Schur moment stays at 1 — irreducible — at both new grades). The non-crossing/Temperley–Lieb structure constants are not a grade-four curiosity; they are the tower's spine.

The parity law, measured and proven in the same session. Grade five (odd weight) has *no* DC channel — there is nothing to occupy; its sheet is all-$\Gamma_{\mathbb{C}}$, $Q = (11 \cdot 37 \cdot 53 \cdot 61 \cdot 79)^{16}$, and the component rules give $\varepsilon = -1$: *a central zero is forced on the sheet* — the pre-committed falsifiable statement that replaces the missing DC reading on odd rungs. Grade six restores the DC channel (20 balanced lanes of 64; middle split 10/10 forced; $\varepsilon = +1$; non-critical center exactly as grade four). The same fact is a theorem the same day (`EvenWeightDC.lean`, §10). Falsifiers at both grades: wrong-harmonic $139\sigma/132\sigma$; scrambling the *degenerate* alignment lifts the frozen channels by 437σ (channel constancy is genuine per-prime alignment, not a moment coincidence), while scrambling the primitive changes nothing (the moments are marginal–Sato–Tate facts, as they must be).

The tower to grade twenty, and the root-number law. The atlas was then run to grade twenty (the first twenty prime-conductor curves; the 2^g-lane enumeration replaced by an exact $O(g^2)$-per-prime polynomial convolution — an algorithm, not a method), and the empirical field produced a theorem-shaped law. The archimedean assembly closes to $\varepsilon(g) = i^{A(g)}$ with

$$A(g) \;=\; 2^{g-1} \;+\; g\binom{g-1}{\lfloor (g-1)/2 \rfloor},$$

always even — so the tower is self-dual with a real sign at *every* grade — and the sign is -1 exactly when $g = 2^k + 1$: by Kummer's theorem the 2-adic valuation of the central binomial is the binary digit sum, so among odd grades only $m = (g-1)/2$ a power of two survives. *Forced central zeros at exactly* $g = 3, 5, 9, 17, 33, \ldots$ — reproducing grade three (the Ceresa channel's forced zero, §8) and grade five (§13), while $g = 7, 11, 13, 15, 19$ are odd with $\varepsilon = +1$, so "odd forces a zero" is false and the law is sharper; grades nine and seventeen stand as new pre-committed forced-vanishing predictions. Alongside it, three more measured laws: the Catalan spine holds exactly to $C_{10} = 16796$ and $C_{20} = 6.56 \times 10^9$; the DC lane fraction falls as $\binom{g}{g/2}/2^g \sim \sqrt{2/\pi g}$ (the homelessness curve, .500 → .176); and the interior Gram of the middle block continues the grade-four $\mathbf{1} \oplus \mathbf{2}$ split as *trivial* $\oplus$ *nontrivial* S_g*-irreps* at every even grade mapped ($\mathbf{1} \oplus \mathbf{9}$ at six, $\mathbf{1} \oplus \mathbf{20} \oplus \mathbf{14}$ at eight; the balanced-bipartition object, the faithful continuation of grade four's pair-directions) — one simple common mode, and an exotic-candidate complement of known representation type. Resolution itself obeys a three-tier law at fixed prime count: bounded channel moments stay flat ($< 1.5\sigma$ through grade twenty); single-clock power moments degrade polynomially; g-clock product readouts (the Schur/primitivity trace) die exponentially at $g \approx 17$ — *the cliff is a property of the readout, not the object*, and the channel decomposition still reads cleanly where the global trace is blind. Finally the falsifier column returned a method law of its own: a probe of fixed support dilutes ($145\sigma \to 83\sigma$ up the tower, *including* a single leg warped to the top harmonic $m = g$ — the dilution axis is the corrupted-leg *count*, not the multiplier), while constant-fraction support holds and grows ($145\sigma \to 493\sigma$). Two lessons, one per axis: within the warp-based falsifier family the dilution law is a *support* law (the corrupted-leg count, not the multiplier), while the carrier-side harmonic *scale* (the μ_6 cells) stays grade-invariant. Method correction, adopted after these runs: falsifier probes belong on the *reading scale* — hold the fiber untouched and move the instrument off the μ_6 grid, on which the object closes and off which it fails — and the warp-based family used here is deprecated for future instruments (its measurements stand as the family's characterization). The corrected family already has a height-side instance on record: the unit-scale window misread of §12. Conductor toll at grade twenty: $\log_{10} Q \approx 2.15 \times 10^7$ — the value chart is twenty-one million digits deep while the structure tier reads on. The symmetric-power side supplies the companion sign law and a constituent-level validation of the same component rules: for semistable E the known closed form [33, 34] — $w_m = +1$ for even m; $w_m = \left(\frac{-2}{m}\right) w(E)$ for odd m — is re-derived honestly from Tate's twisted-Steinberg factor ($\varepsilon_p(\mathrm{Sym}^k) = w_1(p)^k$; no even-power shortcut exists on the Sym side) and confirmed against the PARI oracle on all eighteen signs of $\mathrm{Sym}^{1..9}$ of $11a1$ and $37a1$, with the forced vanishings verified as actual central zeros where values are reachable. And the grade-nine consistency identity closes: the degenerate $V^{\otimes 9} = \bigoplus_k m_k \,\mathrm{Sym}^k$ sheet's diamond sign equals the constituent product $\prod_k \varepsilon(\mathrm{Sym}^k)^{m_k} = -1$ (with $\sum_k m_k A(\mathrm{Sym}^k) = 886 = A$ exactly and Tate twists sign-neutral) — the degenerate shadow of the $g = 2^3 + 1$ forced zero, the two towers' 2-adic sign laws (binary digit sum; Kronecker $\left(\frac{-2}{m}\right)$) imaging each other. The atlas's harmonic compatibility was then audited rather than assumed (`harmonic_compat.py`): the bank is conjugate-closed to machine zero at grades 6, 12, and 20, and every channel moment is *affine* in the single-clock closure value $c(w) = \mathbb{E}[\cos 2w\theta]$ — at the fiber's own fundamental, $c = -\frac{1}{2}$ and the motive's exact rationals close; at any other rational multiplier, $c = 0$ and the reading is a different, decoupled object ($> 100\sigma$ from the motive); at an incommensurate multiplier, c is irrational and every moment goes transcendental — the false-null mechanism of the standing method law, now measured in quantitative form. (One certification pitfall caught in-run: best-fit rational approximation "certifies" fake rationality for incommensurate c; the honest discriminator is lattice membership $c \in \{-\frac{1}{2}, 0\}$.)

The tiered picture. What scales freely is *structure*: sheet, sign, channels, interior geometry, primitivity, occupancy — cost 2^g per prime, method fixed. What recedes doubly-exponentially is the *chart*: $Q = (\prod N)^{2^{g-1}}$ prices center values out at every grade past the degree-six rung, and individual zeros with them — which is why the value-registration law (§11) is the single named gate, at every grade at once, rather than one wall per grade. The classical pattern — one new integral representation per grade, supply exhausted at three — was a chart artifact; the carrier needs two parity templates (heights on odd rungs, regulators on even), and grades one through six now instantiate both.

The pairing law, staked. Program register, with its falsifiers. The session's results assemble into one candidate law: *the central data of* Λ *is the Gram data of a single Hermitian pairing on the double helix*. Its analytic face is the rail-matching law (only conjugate-closed — admissible — objects pair; natural boundaries are single-rail artifacts); its arithmetic face is the drift pairing (jets = pairing determinants: heights on odd rungs, regulators on even — the parity alternation, DC-level Lean-proven); its local decomposition is tropical (reduction-graph Green's functions) + archimedean, per Zhang's admissible theory. Named instances: Tate at grade zero, Birch–Swinnerton-Dyer/Gross–Zagier at one, Beilinson–Flach at two, Gross–Schoen/Zhang/Yuan–Zhang–Zhang at three — certified here on the Klein quartic (§8) — the pinned sheet at four, the structure tier at five and six. Falsifiers, pre-committed: a grade-five central non-vanishing (the sheet forces zero); a grade-one place-by-place height decomposition that fails to close on the certified regulators; a grade-four matching-law assembly whose ledger and tropical sides disagree beyond the archimedean remainder. The law is stated to be shot at — and the grade-one shot has already been fired and missed: on the certified census the canonical height decomposes place-by-place exactly — $\hat{h}(P) = 2\big(\lambda_\infty + \sum_p \frac{n_p}{2} B_2(\{m/n_p\}) \log p\big)$, the single calibrated constant being the classical factor two between the canonical height and the Néron sum — blind on $53a1/61a1$ to 10^{-7} (the house 4^{-n} limit converges *upward to* the prediction, so the residual is truncation, not error), non-circularly against the incomplete-Γ jet to 3.5×10^{-13}, with the rank-two pairing matrix decomposing *entrywise* (regulator = determinant to 4.3×10^{-11}) and the wrong-graph control missing by exactly $\frac{2}{3} \log 37$. Every I_1 tropical share is $\frac{1}{6} \log p$ on the nose (`matching_law_g1.py`, App. A).

14 The terminus: the Hodge conjecture as source exhaustion

The program's end goal is stated plainly, with its logic and its price.

In the carrier's coordinates a rational (p,p)-class is a state with the *Hodge signature*: its clock/weight datum sits in the self-paired, phase-balanced position the ledger assigns to type (p,p). The Hodge conjecture [2, 4] asserts such a class is algebraic — in the register of this program:

$$\boxed{\text{every Hodge-signature state has an algebraic source.}}$$

This is a source-exhaustion statement, the same genre as the carrier's proven exhaustion theorems — every vanishing event has a source; no layer is silent — now asserted at the top of the cycle tower. The two-question test the program applies to every target passes: nothing here assumes the conjecture (the towers, poles, and heights are read unconditionally from point counts and completed values), and nothing is circular (the falsifiers below are concrete). Three facts give the terminus its footing:

1. *The divisor case is the template.* Lefschetz $(1,1)$ [3] is the classical grade-one instance: the Hodge-signature condition detects exactly the algebraic classes, and the carrier's depth-one row (§6) reads that grade's full arithmetic — order, regulator, obstruction — as ledger coordinates.

2. *The pole reading is Tate-shaped.* The carrier reads algebraic-cycle count as DC/pole data natively (`rank_is_dc_residue`; the Sym^2 excess-pole detection of the CM Hodge class is a measured instance of exactly the pole-order reading Tate's conjecture [5] assigns to algebraic classes). The program's detection direction — Hodge signature $\Rightarrow$ some T_d fires — is thus already exercised.

3. *The exhaustion principle has never failed a test.* Every class put to it has appeared at finite depth $(1, 1, 1, 2, 3)$, with the predicted delayed signatures; deliberate search for a phantom has so far produced none.

What the terminus still needs — named exactly, the honest spine of the program:

1. *The transcendental retained coordinate.* Two varieties can share their zeta/Satake data yet differ in Chow and Griffiths groups, so the Satake ledger alone cannot separate the transcendental layer in general. The program must exhibit the additional retained coordinate — the non-archimedean height channel or the ℓ-adic Ceresa class of §8, carried on the ledger's radial/angle channels beyond the weight multiset — and prove it is genuinely retained by the descent, not synthesized after the fact. This is the crux, and we state it as the program's hardest single obligation.

2. *The converse leg: from firing coordinate to cycle.* Detection is half the conjecture; the other half constructs the algebraic source from the fired coordinate — the cycle-theoretic analogue of the converse theorem that turns the main paper's nice L-functions into automorphic representations. The functoriality program of [1] supplies the model: identify the complete invariant package (there: functional equation, continuation, boundedness, exhaustion; here: the graded tower record with its ledger), then invoke or build the recognition theorem that says the package has a unique source. No such recognition theorem exists yet at cycle level; building its carrier form is the program's constructive arm.

The architecture, machine-checked. The reduction is now kernel-fixed (`RequestProject/HodgeDial.lean`; standard footprint): a `HodgeDial` carries the DC condition, rationality, an algebraic-source predicate, and the tower readouts; RETENTION (no silent rational DC mode) and RECOGNITION (fired $\Rightarrow$ sourced) are the two named hypotheses — *asserted by no one* — and `hodge_of_retention_recognition` composes them, unconditionally, into SOURCEEXHAUSTION. On the semisimple model dial, retention is already a theorem (`model_sourceExhaustion_of_recognition` consumes the Vandermonde separation), so there the terminus needs recognition alone. Any future landing of either ingredient, by anyone, assembles into the terminus in the kernel with no further argument; the sentence itself never needs to be said in the author's voice.
Falsifiability register. Pre-committed, in print: the program dies at (a) a *phantom* — a nonzero class silent at every tower depth (one suffices; the carrier radical would be nontrivial on cycles); (b) a *false source* — a Hodge-signature state whose firing coordinates provably admit no algebraic cycle beneath them; (c) a *ledger failure* — a pair of varieties whose full retained ledger (not merely Satake) coincides while their cycle theories differ. The register also demands that the reading

procedures fail when mis-clocked, and this has been exercised deliberately: the wrong root of unity, the wrong tower harmonic, and the wrong harmonic rate each break their readings by orders of magnitude (`harmonic_falsifier.py`), and the root-of-unity ladder μ_3–μ_{12} reconstructs integer point counts exactly with its own roots while missing at the integer level with wrong ones (`mu_ladder.py`). Any hit will be published as prominently as a confirmation. Current count after deliberate searching: zero. The first adversarial instance of (c) has been run and survived: across an isogeny class the 1D readout is provably blind — identical a_p, certified in-house — while the retained ledger separates every pair and redistributes in the exact Cassels-invariant combination (App. A). A second instance has now been run where no shield protects the reading: non-diagonal quartic K3s — the Dwork pencil, genuinely coupled, plus block-separable generics with no symmetry at all, hence no twist characters and no Faltings isogeny shield. All forty-four non-isomorphic pairs separate in exact integer point counts (minimum separation 3992, growing with p); the one forced-isomorphic control pair is identical, as required; and the hardest pair (Dwork $c = 2$ vs $c = 1/2$) collides in its coarse fingerprints and even at individual primes, yet the full trace sequence separates it cleanly — the clause-(c) stress scenario in miniature, and the ledger holds (`k3_nondiag.py`, App. A).

Conclusion

What stands, and at what strength. The depth dictionary is a machine-checked theorem about a non-circular, machine-fixed filtration, transport-invariant, coinciding with Bloch–Beilinson whenever the latter exists. No-silent-layer is unconditional on the semisimple carrier state, and on drift in the model: drift is value-blind, jet-visible, and never silent, with positive-definite response energy — the model form of Beilinson–Bloch anisotropy — and the measured law $k^* =$ drift dimension holds across the census with machine-zero silence. Depth one is measured at machine precision with all three coordinates independently resolved and the obstruction landing on exact squares; depth two fired on schedule; depth three is measured on the Laga–Shnidman family with an exact isotropy boundary and a discovered torsion-translation symmetry, and awaits its from-scratch landing — whose L-side is now on record: the Klein quartic's Ceresa channel carries forced sign -1 and central derivative $0.8299 \neq 0$, matching the proven non-torsion of the cycle (Target 8.1, half landed). Recognition is closed at grade one on house instruments, gate-certified. Grade four — the first unnamed rung — is mapped to its three walls (non-critical center, parity ladder, split no-go) and crossed: the homeless degree-16 fiber is assembled on the carrier, its degenerate case factoring through the lower harmonics exactly, its primitive case certified irreducible, its first datum on record — and its fourth wall, the information wall, is measured on three faces and dismantled: the functional-equation sheet is pinned from the twenty integers with the root number proven $+1$, the strip-interior values exist, the center is read non-vanishing in the framework register, the unread number is identified as a tropical-dominated regulator volume, and the fiber's formal home is a Lean theorem. The instruments that got inside — the harmonic lattice, the conjugate-rail pairing, the portal, the μ_6-address locator — obey one measured admissibility criterion, harmonic self-compatibility, and they scale: grades five and six run with no new method, the Catalan spine and the parity law confirm across the ladder, and the residual obstruction at every grade is one named theorem-target, the value-registration law. The terminus architecture is kernel-fixed with its two hypotheses named and asserted by no one. The end goal is the Hodge conjecture as source exhaustion, and the route to it is the same route the main paper walked for functoriality: build the object, keep the ledger, induce every layer to appear, and let the recognition theorem finish.

A Reproducibility

All instruments are oracle-free — no L-function call inside a locator; published values enter only final verification columns. Lean footprints are uniformly {`propext`, `Classical.choice`, `Quot.sound`}, no `sorry`, audited by `#print axioms`.

- **The filtration dictionary**, `RequestProject/HodgeLedgerFiltration.lean`: `carrierFiltration_antitc` `ledger_kernel`, `mem_carrierFiltration_succ`, `no_early_detection`, `grade_visibility`, `firstVisibleDepth_eq_grade`, `isFirstVisible_unique`, `isFirstVisible_map_iff`, `carrierFiltration_` `blochBeilinson_coincides`, `exhaustive_of_radical_trivial`, `grade_visible_of_nondegenerate`, `radical_of_factorsThrough`, `not_mem_radical_of_detectedPast`.

- **The semisimple theorem**, `RequestProject/CarrierTowerSeparation.lean`: `radical_trivial_of_towe` `momentTower_detects`, `momentTower_radical_trivial`, `momentTower_exhaustive`.

- **The depth-one census**, `tmp/jet_census.py` and `tmp/sha_hinge.py`: full BSD readings at ranks 0–3; $|Ш|$ landed as exact squares $1, 4, 9$ (margins $\leq 3.6 \times 10^{-15}$ at rank zero, $\leq 1.6 \times 10^{-4}$ through rank three); $L^{(r)}/r!$ via incomplete-Gamma Taylor; heights by x-only 4^{-n} doubling; Tate c_p and group-law torsion in-house; Sage-oracle validated once, then run oracle-free.

- **The layer calibration**, `tmp/hodge_layer_calib.py`: order/regulator/obstruction resolved as independent coordinates across the census.

- **The delayed-signature test**, `tmp/tower_exhaustion_test.py`: point-counting tower; CM self-correspondence silent at depth one, first firing at depth two; first nonzero slot matches expected filtration depth for every tested class $(1, 1, 1, 2, 3)$; no class silent at every level.

- **The Griffiths probe**, `tmp/hodge_griffiths_probe.py`: value channel fires for every homologically-trivial cycle in the census; the recursive tower shows classes absent at level one appearing at level ≥ 2.

- **The ledger-failure search, v1**, `tmp/hodge_fast_tests.py`: the $11a$ isogeny class. Identical a_p at every good $p < 400$, certified by point counts — so the 1D readout provably cannot separate the members (Faltings) — while the retained channels separate every pair (Ω, T, c_p, each computed from the curve, never from L), and the Cassels combination $\Omega \prod c_p/|T|^2$ is invariant across the class to 2.2×10^{-16}; $|Ш|$ landed $= 1$ on every member. No ledger failure found. The v2 escalation — a pair with equal *full* ledger and different Ш or, in higher dimension, equal zeta and different Chow data — is the genuine clause-(c) falsifier hunt.

- **The drift model and terminus spine**, `RequestProject/JordanDriftTower.lean` (drift value-blind / jet-visible / never silent; `driftPairing_posDef`; `anisotropic_of_definite_factor`), `RequestProject/HodgeDial.lean` (Retention, Recognition, `hodge_of_retention_recognition`, `model_sourceExhaustion_of_recognition`), `RequestProject/Rung4SplitNoGo.lean` (`split_unit_pairi` All standard footprint, no `sorry`.

- **The harmonic frame, calibrated**, `tmp/hodge_clock_demo.py`: $E_1 \times E_2$ Picard numbers read two ways — period-clock alignments (Hecke scan certified against the exact $j = c_4^3/\Delta$) and count-side moments — 6/6 cases matching Lefschetz $(1,1)$ truth; CM discs $-3, -4$ recovered; the two-CM-fields control correctly refuses to align.

- **The drift ladder**, `tmp/drift_observability.py`: jet ladders $k = 0..4$ for eight curves, root numbers certified in-house (split-vs-direct $\Lambda(3)$); $k^* =$ drift dimension throughout; responses = Reg$\cdot|$Ш$|$; three same-placement falsifier curves (certified by prime discriminant and exhibited integral points) separate cleanly.

- **The dial**, `tmp/twistor_hunt.py`: DC-locking on the product-family twistor dial is discrete and exactly algebraic; irrational controls at $O(1)$; locks are local minima (the base point is a critical point of j — quadratic shoulders).

- **The mis-clocking battery**, `tmp/harmonic_falsifier.py` and `tmp/mu_ladder.py`: wrong root / wrong harmonic / wrong rate each break their readings (separations 10^3–10^7); rungs μ_3–μ_{12} reconstruct integer counts exactly with their own roots (purity at 10^{-15}, DC slots as predicted, sphere-lock at inert primes) and miss at the integer level with wrong ones.

- **Recognition at grade one**, `tmp/heegner_recognition.py`: point counts $\to$ modular image $\to$ Heegner alignment ($D = -7$) $\to$ self-certified lattice (both half-period identities exact) $\to P = (1, 0)$ recognized and verified in exact arithmetic; $\hat{h}(P)/\hat{h}(\text{gen}) = 4.0006$.

- **Depth-three anisotropy**, `tmp/ceresa_anisotropy.py`: $\hat{h}(Q_t)$ over $\mathbb{Q}(\sqrt[3]{t^2-1})$ from scratch; exact torsion boundary at $t = 0, \pm 3$; census non-isotropic 0.59–1.16; the torsion-translation symmetry discovered and group-law-verified in-instrument.

- **Rung four**, `tmp/quadruple_rung.py` (occupancies 0/1/Catalan 2/3 with parity controls), `tmp/weil_detection.py` (the Weil-assembly excess, $\mathbb{Q}$-count 0 vs $\overline{\mathbb{Q}}$ occupancy, orthogonality controls), `tmp/farside_quadruple.py` (the degree-16 fiber: degenerate channel split at 3.3×10^{-16}, coefficient factorization $\zeta^2 * \mathrm{Sym}^{2*3} * \mathrm{Sym}^4$ to 3.6×10^{-14}, Schur irreducibility $0.930 \to 1$ vs $13.703 \to 14$, window growth exponent 0.577).

- **The certified evaluator and value chain**, `tmp/center_reading.py`, `tmp/deg6_center.py`, `tmp/theta_cache.py`: one-contour-pass split identity with two-point split-vs-direct self-certification; gates $\zeta(\frac{1}{2})$, $L(11a1, \frac{1}{2})$; oracle validation PARI `lfunsympow` (degrees 3–5) and Dokchitser (degree 4); the new degree-4 and degree-6 centers; $O(1)$ θ_p caches (148,933 primes to 2×10^6).

- **Wall four, three faces**, `tmp/farside_center.py`, `tmp/carrier_center.py`, `tmp/adapted_center.py`: windowed protocol soundness ($A = 1.5315$ vs analytic 1.5323); the information bound measured (primitive transient $X^{0.19}$, no plateau at any reachable height); solved warps close cells but fail both value gates — summability, not values.

- **The dismantling**, `tmp/wall_dissolve.py`: the sheet from twenty integers (forced 3/3 split), first strip-interior values, mirror zeros, center order zero in the framework register, the tropical layer ($\tau(C_n) = n/12$, $\sum \tau = 2/3$, C_5 moments $m_1 = 0, m_2 = 5/36$).

- **The root number, proven**, `tmp/eps_quadruple.py` and `tmp/eps_quadruple_notes.md`: the Deligne/Tate local computation, nine-for-nine postdiction battery (reduction types recomputed via Tate's algorithm, non-circular), conductor-exponent consistency, Sage oracle cross-checks.

- **The lattice, the rails, the portal, the door**, `tmp/harmonic_lattice.py` (exact lane identities, incommensurate falsifier), `tmp/helix_pairing.py` (single rail diverges / conjugate pair converges to the certified value / both mis-pairs diverge), `tmp/hodge_portal.py` (channel moments, zero-variance degenerate constancy, interior Gram 17/32–1/4, 144σ falsifier), `tmp/spectral_door.py` and `tmp/spectral_door2.py` (fixed-head μ_6-rail locator, 12/12 oracle gate, Rayleigh-limit pair split, quadruple clean null with hostile checks); `tmp/helix_tower.py` spectral half: the growth-window rerun of the gate (11/12 bare rails; the diagonal schedule starves at tensor conductor, as diagnosed) and of the quadruple (pre-registered null; the recurrent $t = 2.68$ dip killed a second time by Z-wander and both scrambles); the arrow-scale falsifier shown to be gauge in fixed-head scans (sign-scramble is the honest falsifier there, 0/12); and the growth-vs-flat resolution comparison — a bounded null: at reachable head the grown and equal-magnitude reads resolve the close pair at the same data volume; `tmp/helix_toll.py` and `tmp/helix_spectral.py`: the $g = 6$ follow-up with two pre-registered predictions falsified and one first-run claim retracted (the transient growth exponent tracks *degree* through phase cancellation, not conductor — a third independent confirmation that cheap descriptors carry no value information; the carrier-representation identity holds at both grades), and the Sym^4 escalation: 8/19 oracle zeros located at degree five, the detected subset head-stable and dying under both scrambles — the locator's first degree-five focal events.

- **The Ceresa L-side**, `tmp/ceresa_lside.py`: from-scratch Hecke L-series over $\mathbb{Q}(\sqrt{-7})$; integer-exact match to the weight-4 level-49 CM form; two-point sign certificates; central derivative 0.8298893 (PARI agreement 3.6×10^{-6}); the model subtlety (naive Klein model is a cubic twist) caught by point counts.

- **The scaling test**, `tmp/grade56_scaling.py`, `tmp/theta_79a1.npy`, `tmp/theta_83a1.npy`: pre-registered exact rationals at grades five and six, all landing $\leq 1.2\sigma$; Catalan spine $C_3 = 5$ and Schur companions C_4, C_5, C_6; parity law (no DC channel at grade five); falsifiers 132–139σ, degenerate-scramble 437σ; design-change log: the odd-weight Γ branch only.

- **The clause-(c) hunt off the diagonal**, `tmp/k3_ledger.py` and `tmp/k3_nondiag.py`: the diagonal-quartic reader (Fermat class $\rho - 1 = 19$ at split primes, exact locks on inert progressions) and the non-diagonal escalation (Dwork pencil and block-separable generics; two independent exact counters validated against brute enumeration; forty-four pairs all separated, forced-isomorphic control identical; exact-discriminant screening catching a bad prime at 229 the small-p screen missed).

- **The tower atlas**, `tmp/tower_atlas.py`: grades 2–20 on the first twenty prime-conductor curves; exact $O(g^2)$ convolution portal (brute-checked through $g = 10$); the $\varepsilon(g)$ law with the Kummer derivation; Catalan spine to C_{20}; three-tier resolution law with theoretical (not empirical) error bands; three-version falsifier sweep with the support-dilution curve; `tmp/harmonic_compat.py`: the compatibility audit (conjugate closure at machine zero, the affine-in-$c(w)$ channel-moment law, rational-vs-incommensurate discrimination by lattice membership); `tmp/helix_tower.py`: the height run (Sym^3 pairing converges on the carrier to the certified value, growth -0.067, single-rail and mis-pair diverge; the degree-16 $G * \bar{G}$ identity at 2.4×10^{-13}; the radial-head toll measured; the unit-scale window falsifier).

- **Recognition at grade four**, `tmp/recognition_loop_g4.py`: the four-arm loop on three quadruples (degenerate / two isogeny pairs / scrambled), truth consulted only at grading; the live-direction law; independent counting- and period-side alignment certificates; isogeny degrees found with 10^{-15}-scale j-residual landings; occupancy and channel-profile recognition certificates; scrambled null and false-alignment rejection.

- **Recognition at grade two (even rung)**, `tmp/bf_loop_g2.py` and `tmp/bf_loop_g2_notes.md`: the trivial-zero/FE bridge derived and confirmed; the certified Rankin–Selberg values at the Beilinson point; the Rogers–Zudilin corner closed from scratch (Mahler integral vs evaluator, 1.55×10^{-5}, rational landing 24); the genuine-product construction gap documented as the literature's own state; the carrier translation of the even-rung construction arm.

- **The value-registration census**, `tmp/value_registration.py`: frozen forcible-closure protocol (faithfulness anchor reproducing the published mis-registrations exactly); 24 rows over the six certified truths; leave-one-object-out cross-validation killing the closure-improvement and degree/conductor candidates and confirming off-lattice distance as the governing regressor; the graded-family monotone degradation ($0.165 \to 0.200 \to 0.228 \to 0.236$ as off-lattice freedom is added); the finite-difference Jacobian requirement (real banks are first-order-stationary for the closure objective) documented in-source.

- **The grade-one pairing-law calibration**, `tmp/matching_law_g1.py`: canonical heights decomposed place-by-place (archimedean theta/sigma via complex AGM — the true lattice $\tau = \omega_2/\omega_1$, not j-inversion's fundamental-domain representative — plus Bernoulli Green's functions on the reduction graphs); one calibrated constant (the classical factor two), then frozen and blind; jet comparison non-circular; wrong-graph falsifier missing by the exact predicted rational.

- **The rung-four Lean bricks**, `RequestProject/QuadrupleFiber.lean` (thirteen theorems: the degree-16 fiber's determinant-one ledger, $(-X)^{16}$ per-place functional equation, completed reflection, strand exchange) and `RequestProject/EvenWeightDC.lean` (ten theorems: `dc_channel_iff_even_weight` and the grade-4/grade-5 corollaries). Standard footprint on all twenty-three, no `sorry`.

References

[1] S. Lavery, *Carrier, fiber, adapters, and functoriality*, preprint draft v0.1 (2026); the 3D carrier system, the loss ledger and its bijection, the functoriality program, and the reproducibility corpus this paper builds on.

[2] W. V. D. Hodge, *The topological invariants of algebraic varieties*, Proc. ICM 1950, vol. 1, 182–192.

[3] S. Lefschetz, *L'Analysis situs et la géométrie algébrique*, Gauthier-Villars, Paris, 1924.

[4] P. Deligne, *The Hodge conjecture*, in *The Millennium Prize Problems*, Clay Math. Inst./AMS (2006), 45–53.

[5] J. Tate, *Algebraic cycles and poles of zeta functions*, in *Arithmetical Algebraic Geometry* (Purdue, 1963), Harper & Row (1965), 93–110.

[6] A. Beilinson, *Height pairing between algebraic cycles*, in *K-theory, Arithmetic and Geometry*, Lecture Notes in Math. **1289**, Springer (1987), 1–26.

[7] S. Bloch, *Height pairings for algebraic cycles*, J. Pure Appl. Algebra **34** (1984), 119–145.

[8] J. P. Murre, *On a conjectural filtration on the Chow groups of an algebraic variety*, Indag. Math. (N.S.) **4** (1993), 177–201.

[9] B. J. Birch and H. P. F. Swinnerton-Dyer, *Notes on elliptic curves. II*, J. reine angew. Math. **218** (1965), 79–108.

[10] B. H. Gross and D. B. Zagier, *Heegner points and derivatives of L-series*, Invent. Math. **84** (1986), 225–320.

[11] V. A. Kolyvagin, *Euler systems*, in *The Grothendieck Festschrift, vol. II*, Progr. Math. **87**, Birkhäuser (1990), 435–483.

[12] S. Zhang, *Gross–Schoen cycles and dualizing sheaves*, Invent. Math. **179** (2010), 1–73; the Beilinson–Bloch height of the modified diagonal.

[13] X. Yuan, S. Zhang, and W. Zhang, *Triple product L-series and Gross–Schoen cycles*, book manuscript; the central derivative of the triple-product L-function as a Gross–Schoen/Ceresa height.

[14] G. Ceresa, *C is not algebraically equivalent to C^- in its Jacobian*, Ann. of Math. (2) **117** (1983), 285–291.

[15] B. Harris, *Harmonic volumes*, Acta Math. **150** (1983), 91–123, and *Homological versus algebraic equivalence in a Jacobian*, Proc. Natl. Acad. Sci. USA **80** (1983), 1157–1158.

[16] J. Laga and A. Shnidman, *Ceresa cycles of bielliptic Picard curves*, arXiv:2312.12965 (2024).

[17] M. Flach, *A finiteness theorem for the symmetric square of an elliptic curve*, Invent. Math. **109** (1992), 307–327.

[18] W. T. Gan, B. H. Gross, and D. Prasad, *Symplectic local root numbers, central critical L-values, and restriction problems in the representation theory of classical groups*, Astérisque **346** (2012), 1–109.

[19] W. Zhang, *Weil representation and arithmetic fundamental lemma*, Ann. of Math. (2) **193** (2021), 863–978.

[20] C. Voisin, *Some new results on modified diagonals*, Geom. Topol. **19** (2015), 3307–3343; the vanishing $\Gamma^{g+2}(C, x) = 0$.

[21] P. Deligne, *Les constantes des équations fonctionnelles des fonctions L*, in *Modular Functions of One Variable II*, Lecture Notes in Math. **349**, Springer (1973), 501–597.

[22] J. Tate, *Number theoretic background*, in *Automorphic Forms, Representations and L-functions* (Corvallis), Proc. Sympos. Pure Math. **33.2**, AMS (1979), 3–26.

[23] D. E. Rohrlich, *Elliptic curves and the Weil–Deligne group*, in *Elliptic Curves and Related Topics*, CRM Proc. Lecture Notes **4**, AMS (1994), 125–157.

[24] Y. Tadokoro, *A nontrivial algebraic cycle in the Jacobian variety of the Klein quartic*, Math. Z. **260** (2008), 265–275.

[25] N. D. Elkies, *The Klein quartic in number theory*, in *The Eightfold Way*, MSRI Publ. **35**, Cambridge Univ. Press (1999), 51–101.

[26] T. Estermann, *On certain functions represented by Dirichlet series*, Proc. London Math. Soc. (2) **27** (1928), 435–448.

[27] N. Kurokawa, *On the meromorphy of Euler products I*, Proc. London Math. Soc. (3) **53** (1986), 1–47.

[28] A. Beilinson, *Higher regulators and values of L-functions*, J. Soviet Math. **30** (1985), 2036–2070.

[29] A. Lei, D. Loeffler, and S. L. Zerbes, *Euler systems for Rankin–Selberg convolutions of modular forms*, Ann. of Math. (2) **180** (2014), 653–771.

[30] M. Bertolini, H. Darmon, and V. Rotger, *Beilinson–Flach elements and Euler systems I*, J. Algebraic Geom. **24** (2015), 355–378.

[31] F. Brunault and M. Chida, *Regulators for Rankin–Selberg products of modular forms*, Ann. Math. Qué. **40** (2016), 221–249.

[32] M. Rogers and W. Zudilin, *On the Mahler measure of* $1+X+1/X+Y+1/Y$, Int. Math. Res. Not. (2014), 2305–2326; and *From L-series of elliptic curves to Mahler measures*, Compositio Math. **148** (2012), 385–414.

[33] N. Dummigan, P. Martin, and M. Watkins, *Euler factors and local root numbers for symmetric powers of elliptic curves*, Pure Appl. Math. Q. **5** (2009), 1311–1341.

[34] T. Saito, *The sign of the functional equation of the L-function of an orthogonal motive*, Invent. Math. **120** (1995), 119–142.